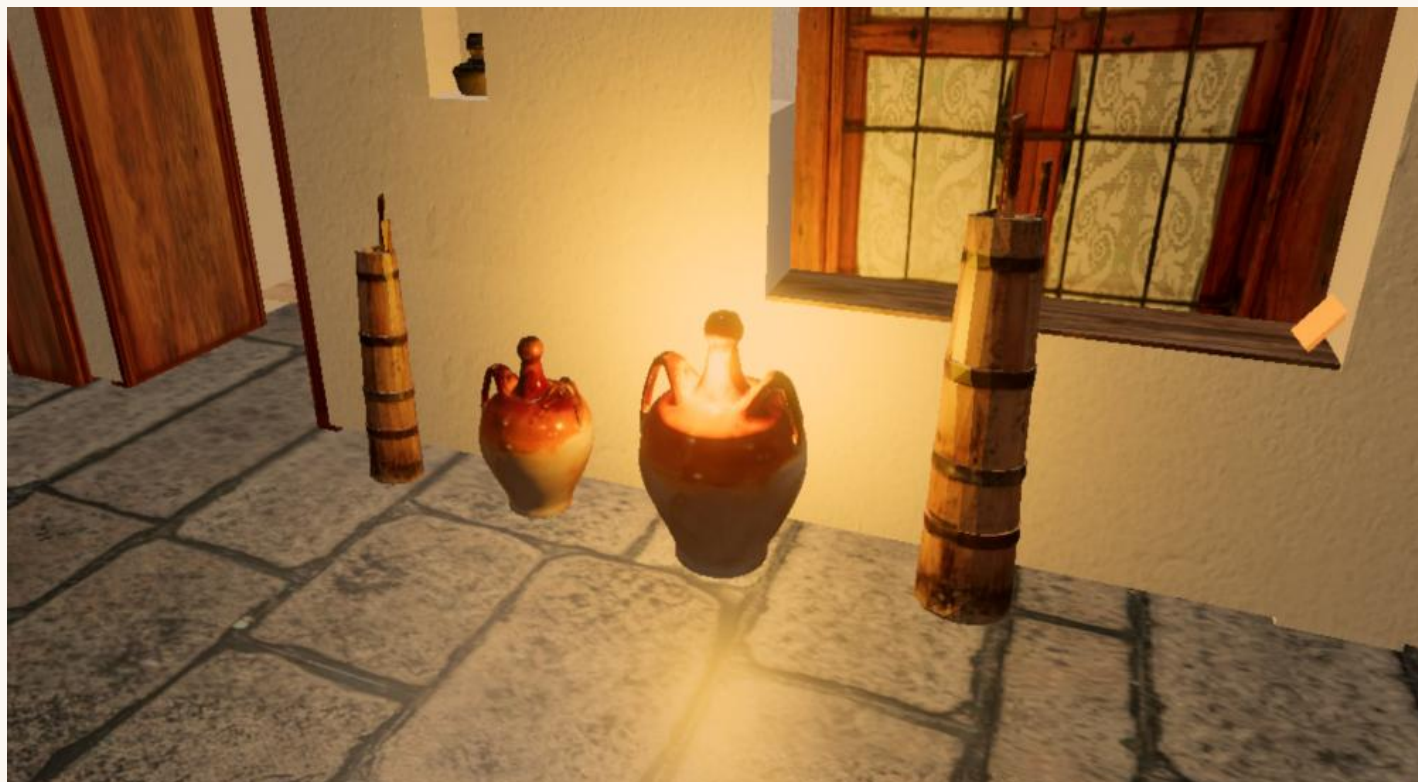


Adaptive Virtual Reality Museum

A Closed-Loop Highlight Intervention Study in Cultural-Heritage VR

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Reconstructed rooms of the Berat Ethnographic Museum (Albania) inside the VR experience.

Background & Motivation

- Static VR museums treat every visitor the same way, yet modern stand-alone headsets stream rich behavioural signals (gaze, head, motion, hand) that could reveal exactly when a visitor is losing interest. We use those signals to gently redirect attention without interrupting the experience.

Goal

- Test whether engagement-driven timing of a visual highlight cue improves cultural-heritage exploration in VR, compared with a fixed-schedule version of the same cue, with everything else held constant.

Core Idea

- A closed-loop pipeline reads the visitor's behaviour twice per second, classifies engagement into five levels, and, only when needed, gently glows a nearby artefact the visitor has not yet examined, redirecting their gaze back into the museum.

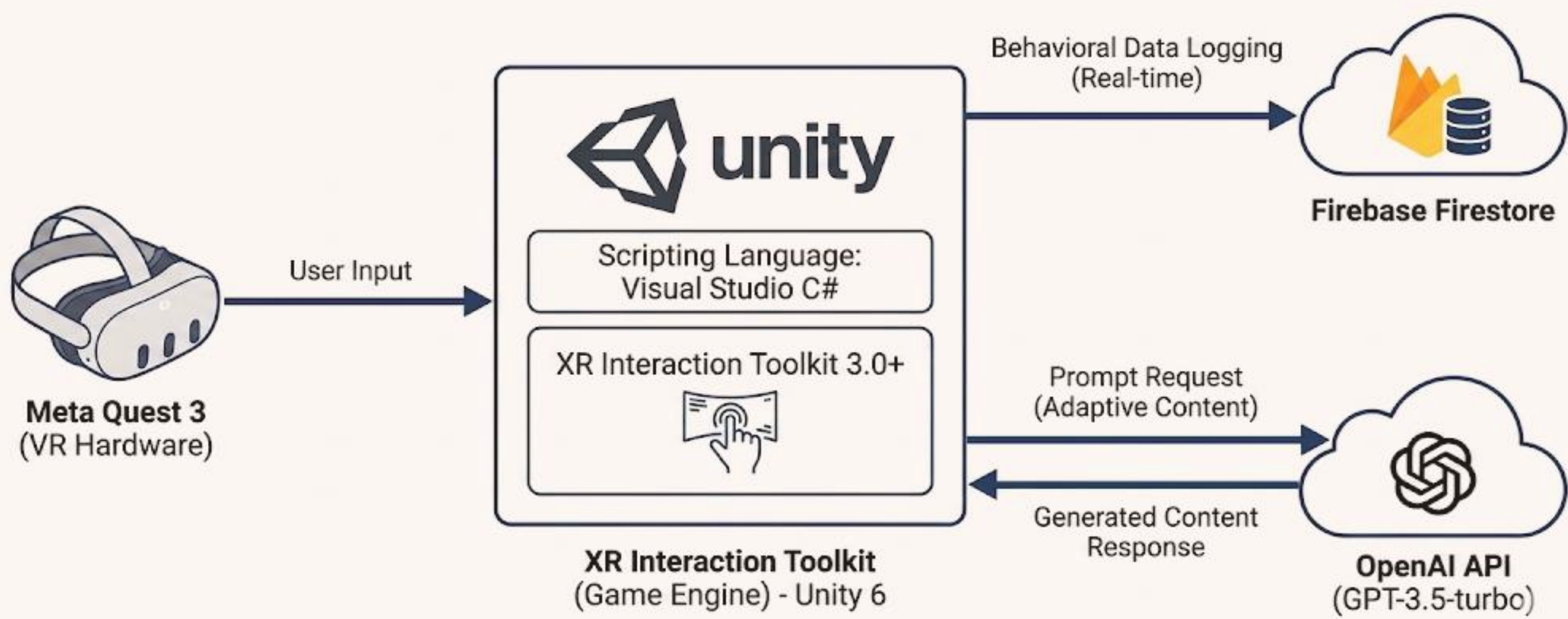
System Objectives

- Reconstruct the Berat Ethnographic Museum as a six-room VR scene on Meta Quest 3.
- Infer a five-level engagement state on-device, every 2 s.
- Deliver a transient perceptual highlight on a chosen artefact when the visitor would benefit.
- Persist every behavioural sample and event to Firebase for reproducible analysis.
- Stay accessible: bilingual UI (English + Albanian), gamification, stand-alone deployment.

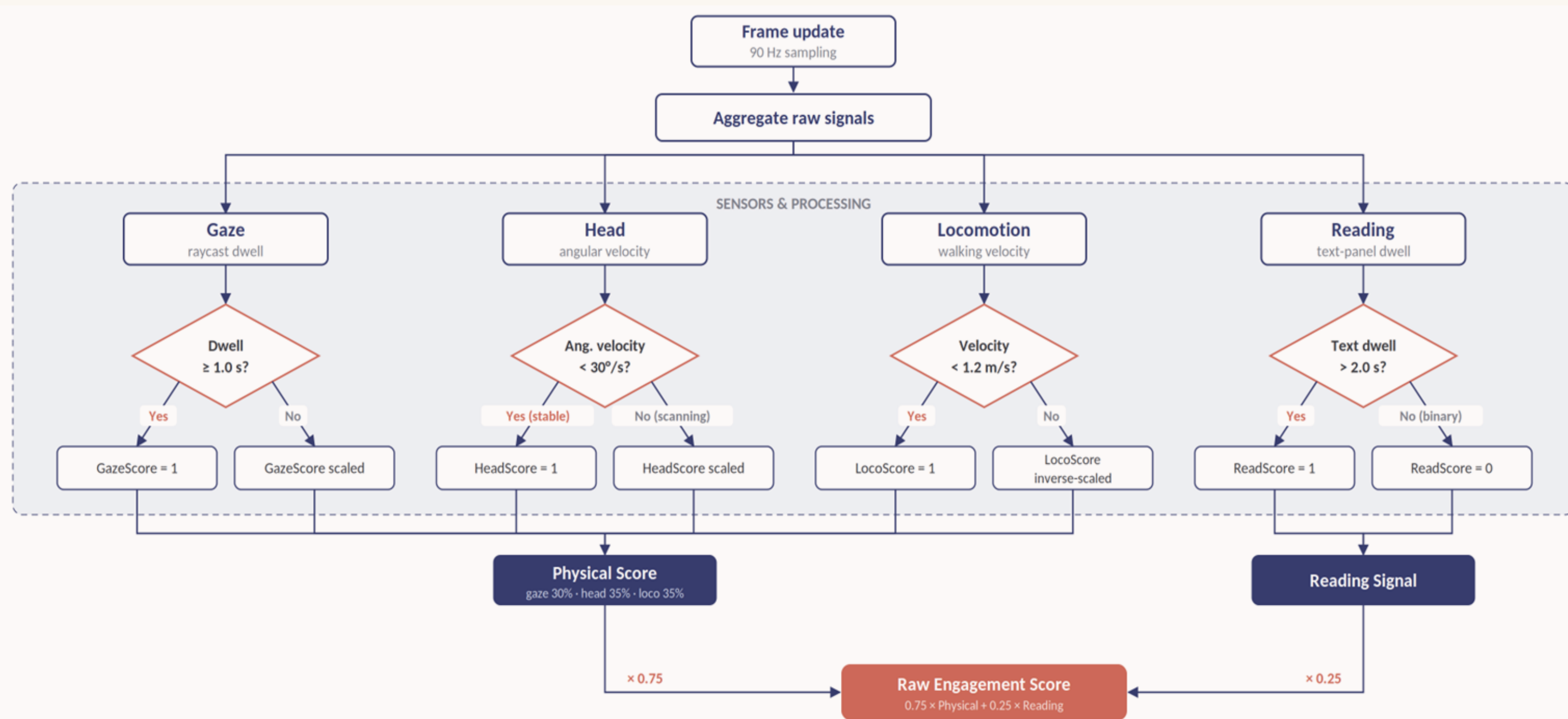
Technologies & Tools

Hardware Meta Quest 3 (Snapdragon XR2 Gen 2)
Engine Unity 6000.2.8f1
VR SDK OpenXR · XR Interaction Toolkit · Meta XR
Language C#
Backend Firebase Firestore + Auth
AI OpenAI GPT

Architecture



Engagement Model Flow



Process Workflow

Foundation

1

Build the museum

3D reconstruction of the Berat Museum: six rooms, photogrammetry + Blender, English & Albanian.

Prototype

2

Adaptive text length

Engagement classifier + GPT adapt exhibit text length. Pilot N = 16; paper accepted at IUI 2026.

Redesign

3

A stronger direction

Text-length change had only a small effect. Pivot: instead of changing what visitors read, the AI now guides where they look, via a gentle artefact glow.

Phase B Pilot

4

Engagement-driven cue

Two-group pilot: engagement-driven glow vs the same glow on a fixed timer. Same headset, scene, and visual.

Results

Metric	Adaptive M (SD)	Control M (SD)	p	effect
Session duration (s)	393 (66)	400 (188)	0.943	$g = -0.05$
Reading / content view time (s)	104 (44)	95 (66)	0.830	$g = 0.14$
Content opens (count)	18.1 (7.9)	13.2 (8.9)	0.517	$g = 0.55$
Highlight dose (count)	2.8 (1.3) ‡	6.0 (2.8)	0.064	$g = -1.28$
Follow-through (acted on)	87% ‡	35%	0.066	$g = 1.26$
Gaze follow-through	87%	55%	0.227	$g = 0.75$
Latency to interaction (s)	5.6 (1.4) ‡	2.8 (2.3)	0.233	$g = 1.38$
H3 recovery / session (Δ)	0.196 (0.159) ‡	-0.038 (0.076)	0.036	$g = 1.79$

Table 2. Behavioral outcomes by condition. M (SD); p = Mann–Whitney U; effect = Hedges g. ‡ = bootstrap 95% CI of the difference excludes 0.

Highlight Effectiveness

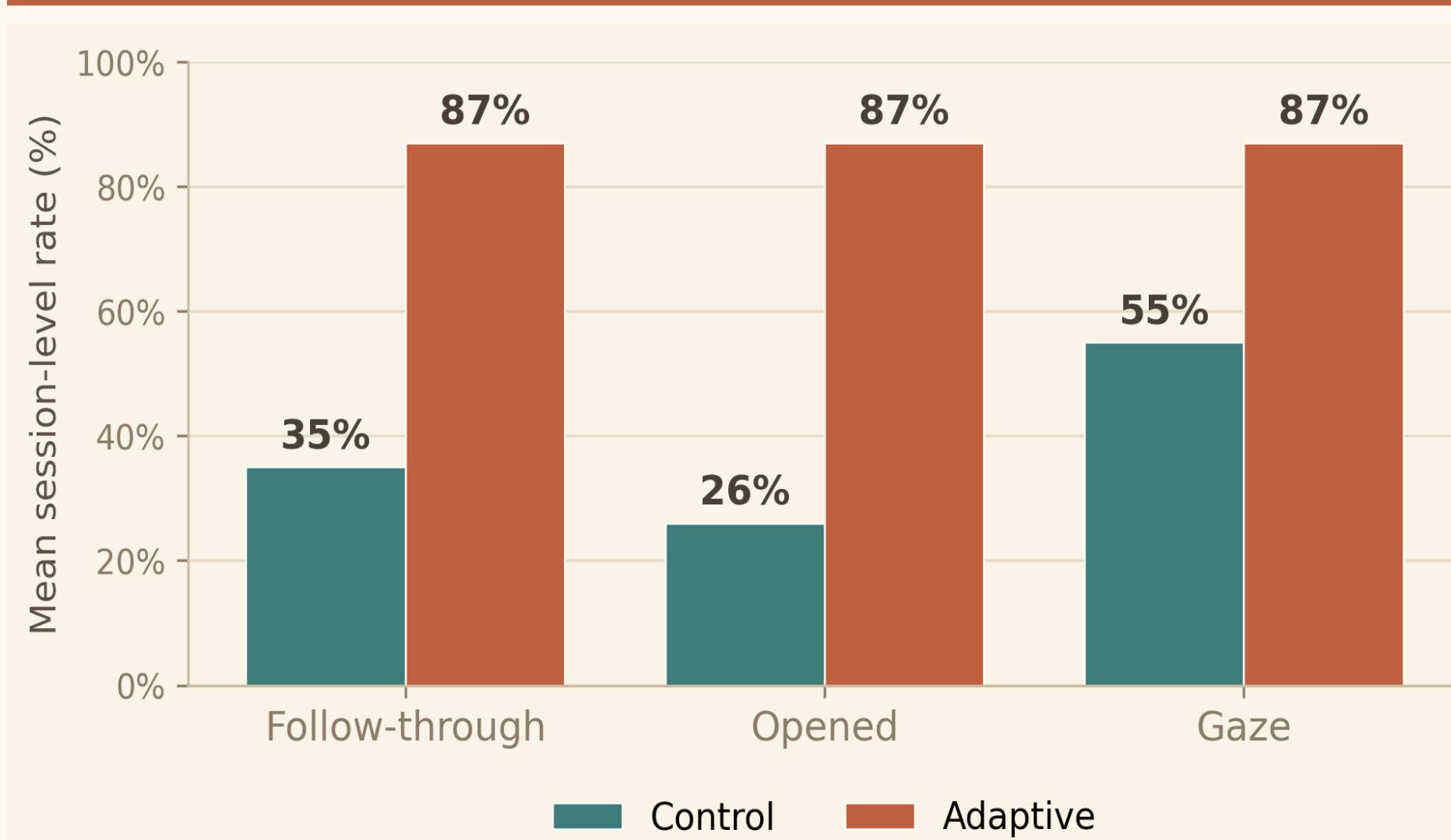


Figure 2. Follow-through, opened, gaze (live-tracker). Exploratory.

CHALLENGES

- Head-tracker bug froze logging in 3 sessions - fixed mid-study
- Reading-protection gate suppressed some Adaptive cues
- Small exploratory pilot (N = 13) - directional evidence only

PUBLICATIONS

- [1] J. Damouni, W. Tanus, and N. Unkelos-Shpigel, "Adaptive Virtual Reality Museum: A Closed-Loop Framework for Engagement-Aware Cultural Heritage," SHAPEXR Workshop at ACM IUI 2026, Paphos, Cyprus, 2026.
- [2] J. Damouni, W. Tanus, and N. Unkelos-Shpigel, "Adaptive Virtual Reality Museum: A Closed-Loop Framework for Engagement-Aware Cultural Heritage," arXiv preprint arXiv:2603.13639, 2026.
- [3] N. Unkelos-Shpigel, W. Tanus, and J. Damouni, "Adaptive Content Pacing in a Virtual Reality Museum: A Preliminary Evaluation of LLM-Driven Personalization," AVI-CH Workshop at AVI 2026, Venice, Italy, 2026.
- [4] J. Damouni, W. Tanus, and N. Unkelos-Shpigel, "Adaptive Engagement Recovery in a VR Museum: An Engagement-Driven Highlight Policy," submitted to IEEE ISMAR 2026 (poster), under review.